Exploratory Data Analysis

Dataset: AI Assistant Usage in Student Life
Notebook Link: AI Assistant Usage in Student Life.ipynb
Dataset:https://www.kaggl
e.com/datasets/ayeshasal89/ai-assistant-usage-in-student-life-synthetic

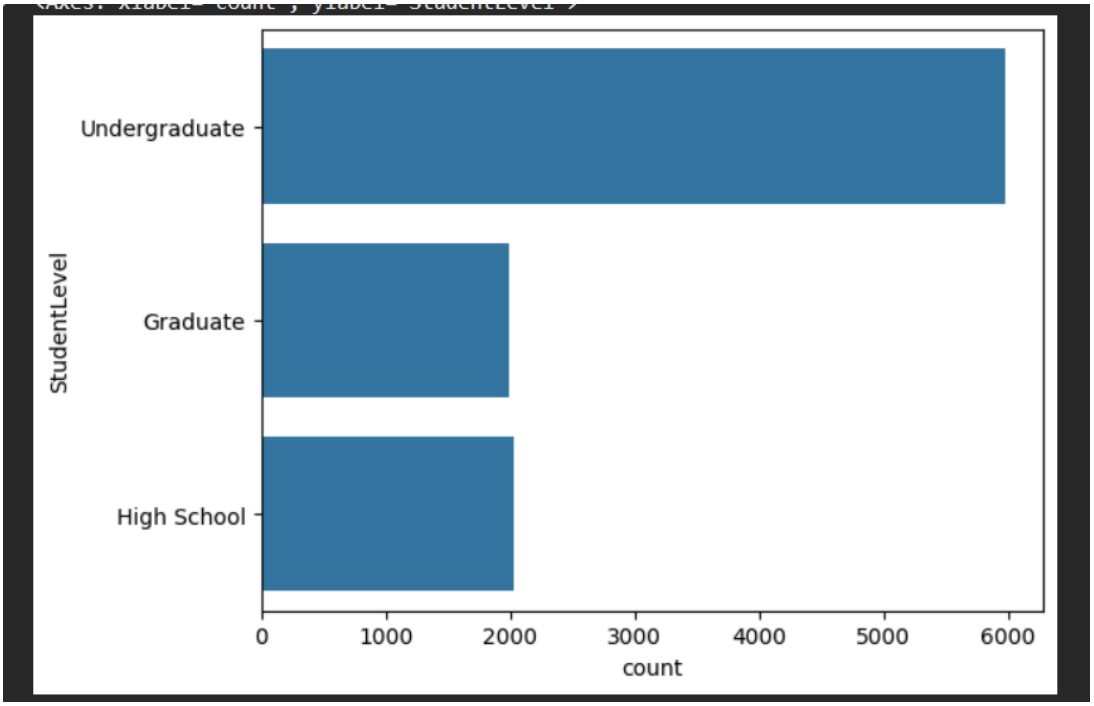
About Dataset This dataset simulates **10,000 sessions** of students interacting with an AI assistant (like ChatGPT or similar tools) for various academic tasks. Each row represents a single session, capturing the student's level, discipline, type of task, session length, AI effectiveness, satisfaction rating, and whether they reused the AI tool later.

Columns:

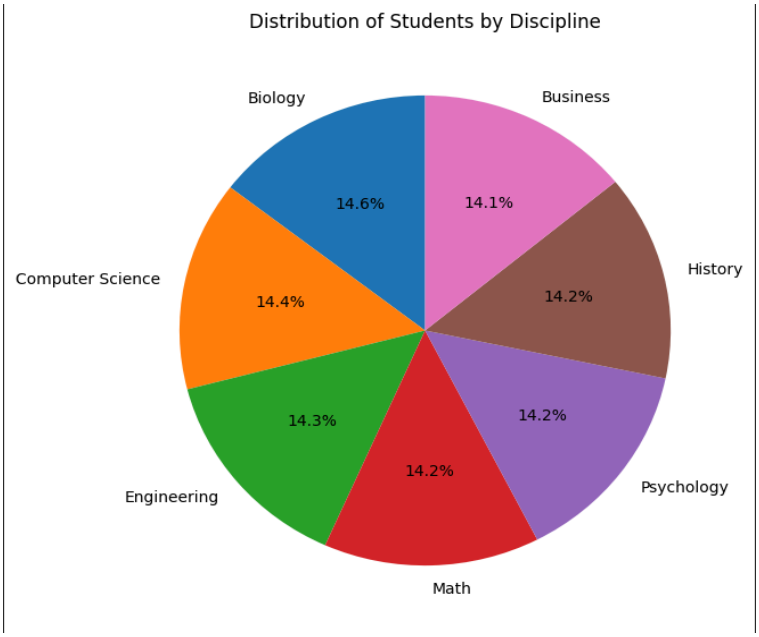
Column	Description
`SessionID`	Unique session identifier
`StudentLevel`	Academic level: High School, Undergraduate, Graduate
`Discipline`	Student's field of study (e.g., CS, Psychology, etc.)
`SessionDate`	Date of the session
`SessionLengthMin`	Length of AI interaction in minutes
`TotalPrompts`	Number of prompts/messages used
`TaskType`	Nature of the task (e.g., Coding, Writing, Research)
`AI_AssistanceLevel`	1–5 scale on how helpful the AI was perceived to be
`FinalOutcome`	What the student achieved: Assignment Completed, Idea Drafted, etc.
`UsedAgain`	Whether the student returned to use the assistant again
`SatisfactionRating`	1–5 rating of overall satisfaction with the session

Insights

1 Mostly Undergraduate students use AI assistant to complete their tasks.



2 The pie chart shows the distribution of students across different fields of study

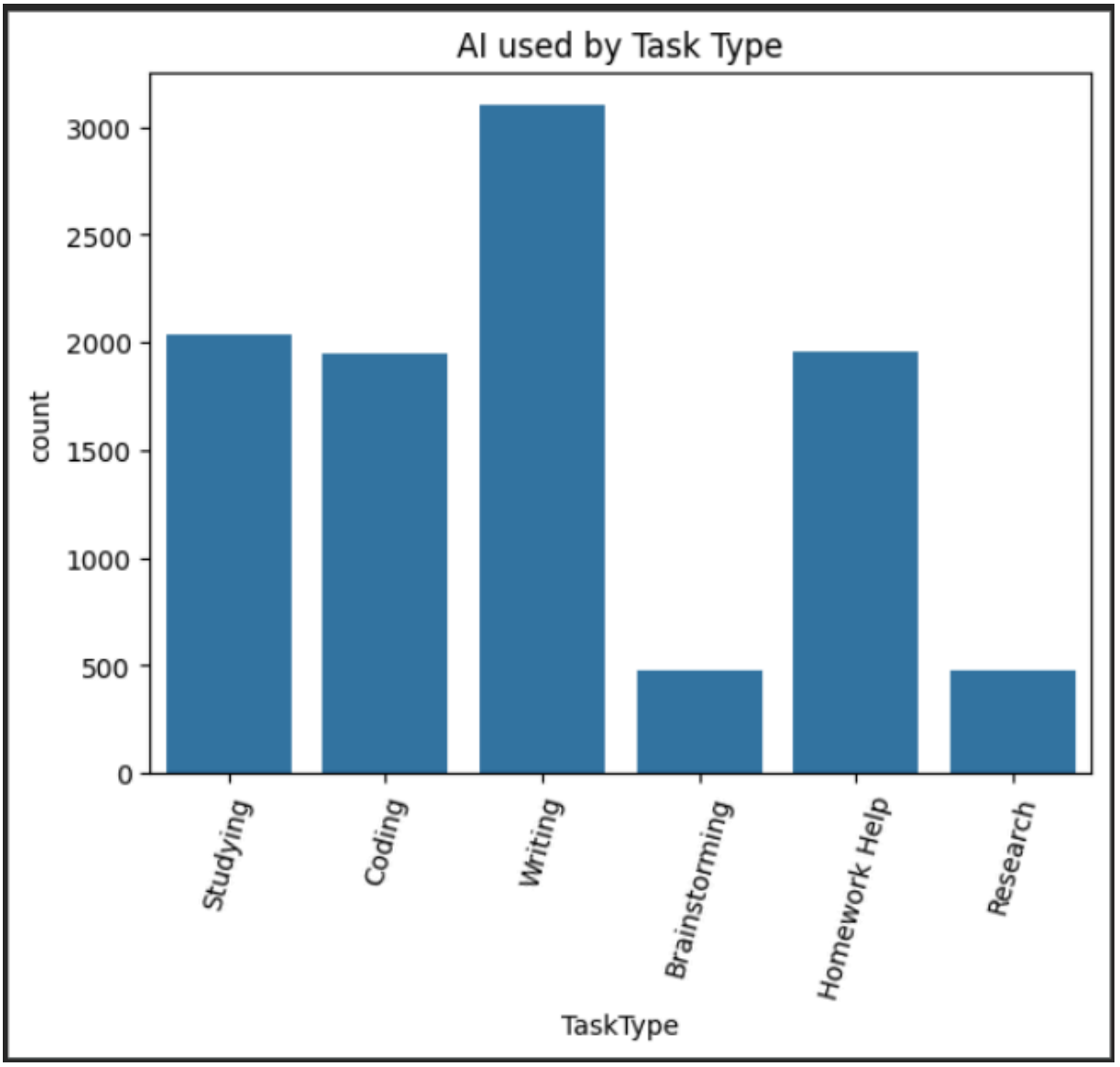


This pie chart clearly show we have more students from

- Biology
- Computer Science
- Engineering
- Math, History and Psychology
- Business

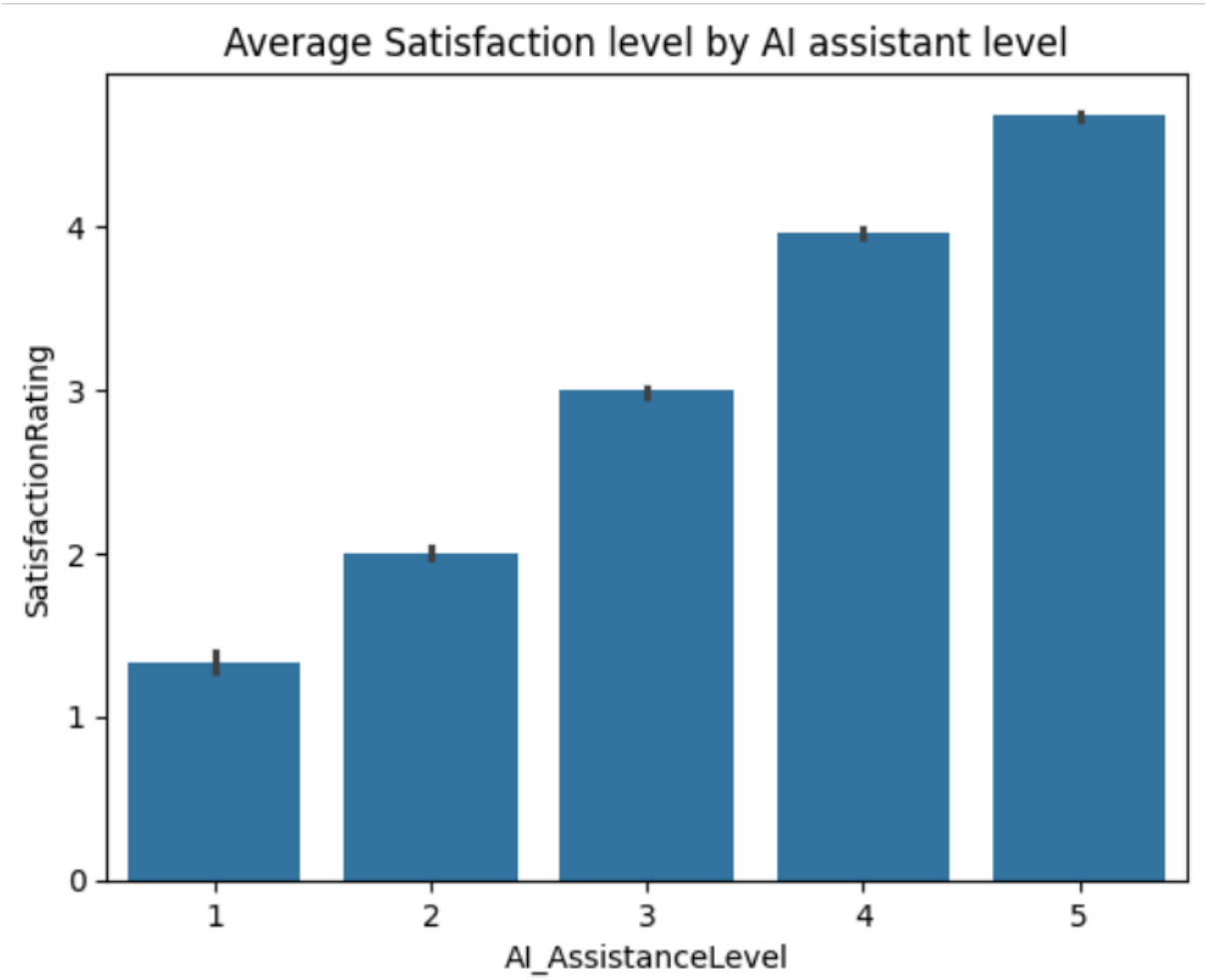
3 In Which task students mostly take help from AI assistants and which level of ai assistant helps them more to complete their tasks.?

The countplot shows that writing tasks are the most frequently completed with the help of AI assistants.



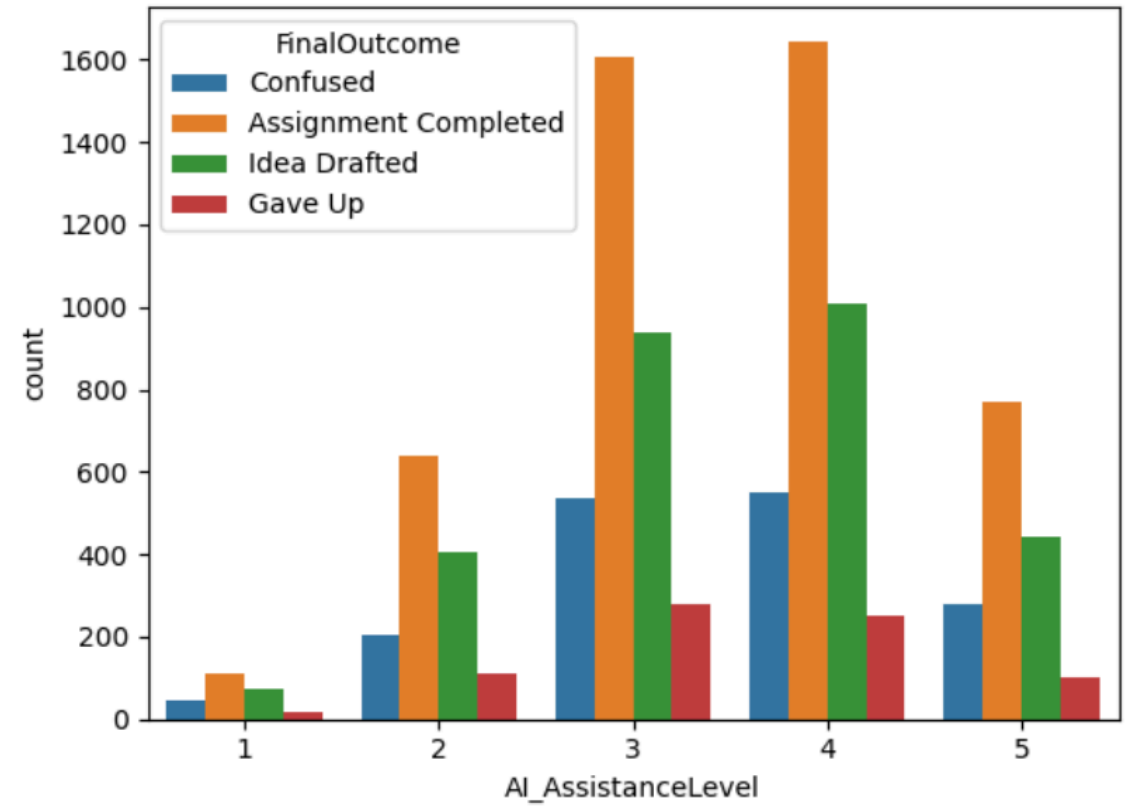
Till now, we saw that large proportion of student comes from Biology and writing is most frequent task, it is likely that many Biology students use AI assistant for writing-related tasks.

4 SatisfactionRating of different AI_AssistanceLevel



AI_AssistanceLevel 5 is highly rated among AI assistance tools.

5 AI Assistant Level (1 to 5) matters in final outcome .

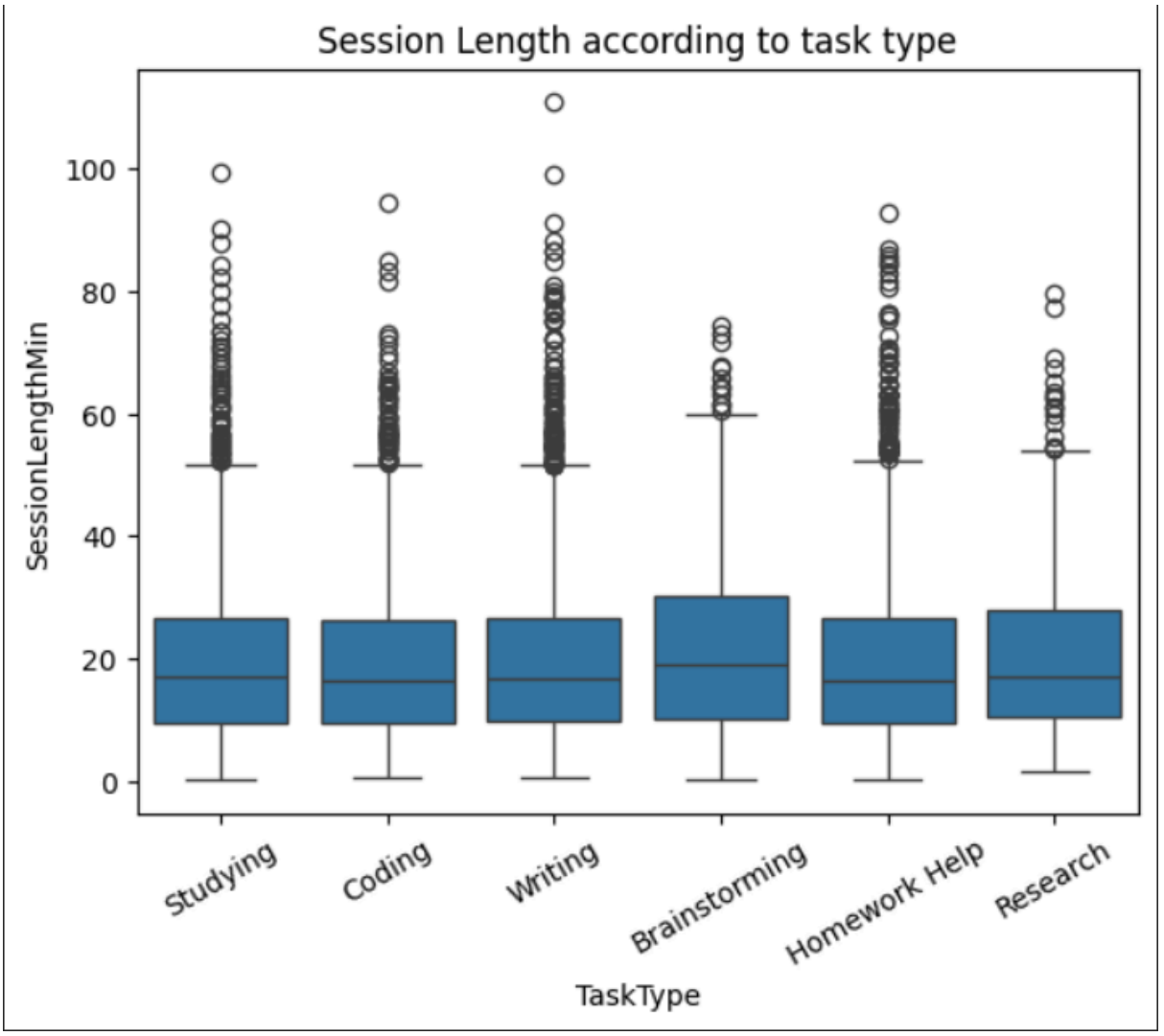


Although AI Assistance Level 4 is used more for completing assignments, the quality of the final outcome is better with AI Assistance Level 3. This is because there are fewer cases of confusion.

From this plot, we can see that the usage of AI Assistance Level 5 is low, but the satisfaction rating is high. So, it might be that AI Assistance Level 5 is paid.

Till now, we saw that Undergraduate Biology students use AI assistants mostly for writing tasks. In this barplot, we can see that the most done task is completing assignments. So, Biology students use AI Assistance Level 4 to help with their assignments.

6 What session length is sufficient to accomplish different academic tasks using AI assistants?



Boxplot shows :

In case of Studying task students spent average time is around 18 minutes and maximum session length goes around 99 minutes

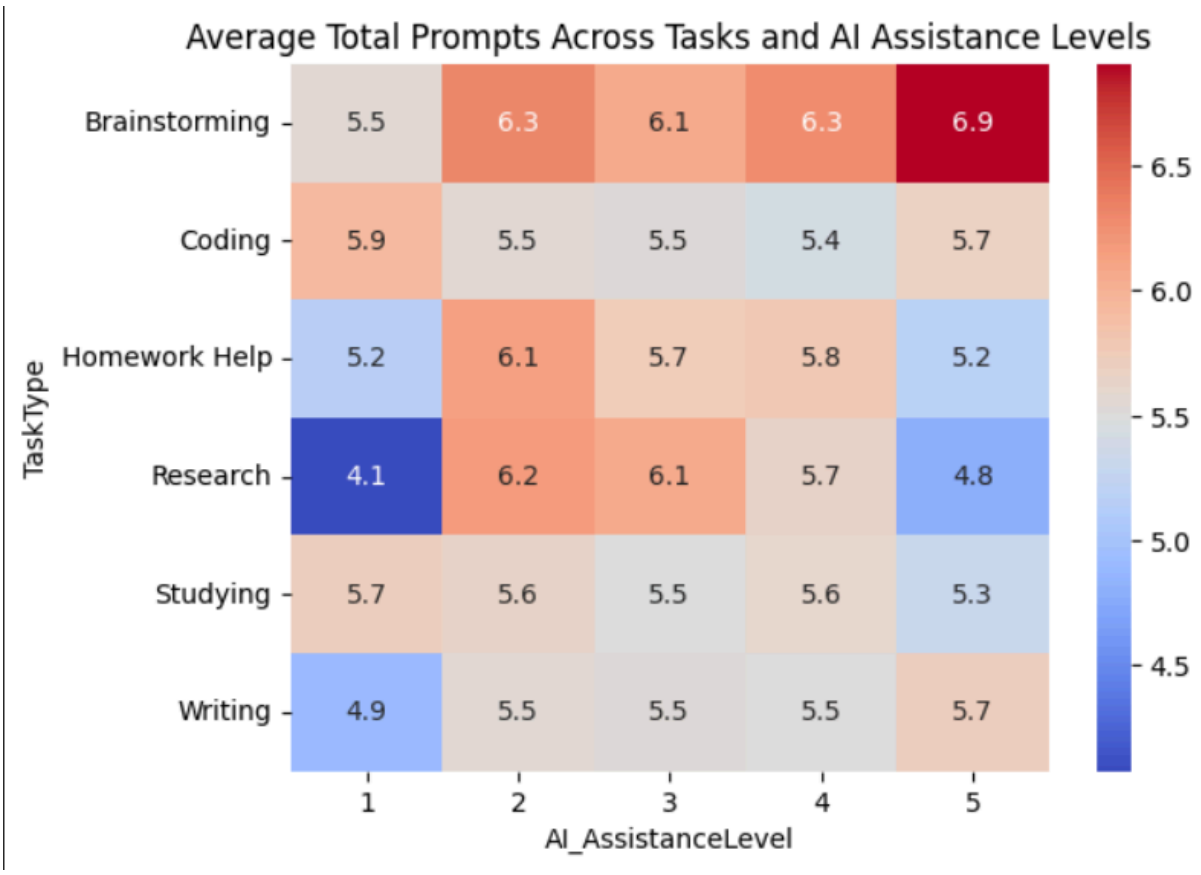
In case of coding average session length is around 17 minutes and maximum session length is around 95.

In case of Writing task students spent average time 18 minutes and maximum time is more than 100 minutes.

For Research average session lengths are around 15 and maximum session length are 80 minutes.

According to this analysis Researchers might use AI assistant level 5 than others.

7 How much total prompts are required to complete task according to AI assistant level?



Numbers inside cells are average count of prompts

Brainstorming task is more demanding.

Different tasks use different AI tools,and higher level assistant level does not mean fewer prompts.

This heatmap shows how many prompts students use for different tasks at different AI assistance levels.

For brainstorming and writing, students use more prompts when the AI level is higher because they explore more ideas.

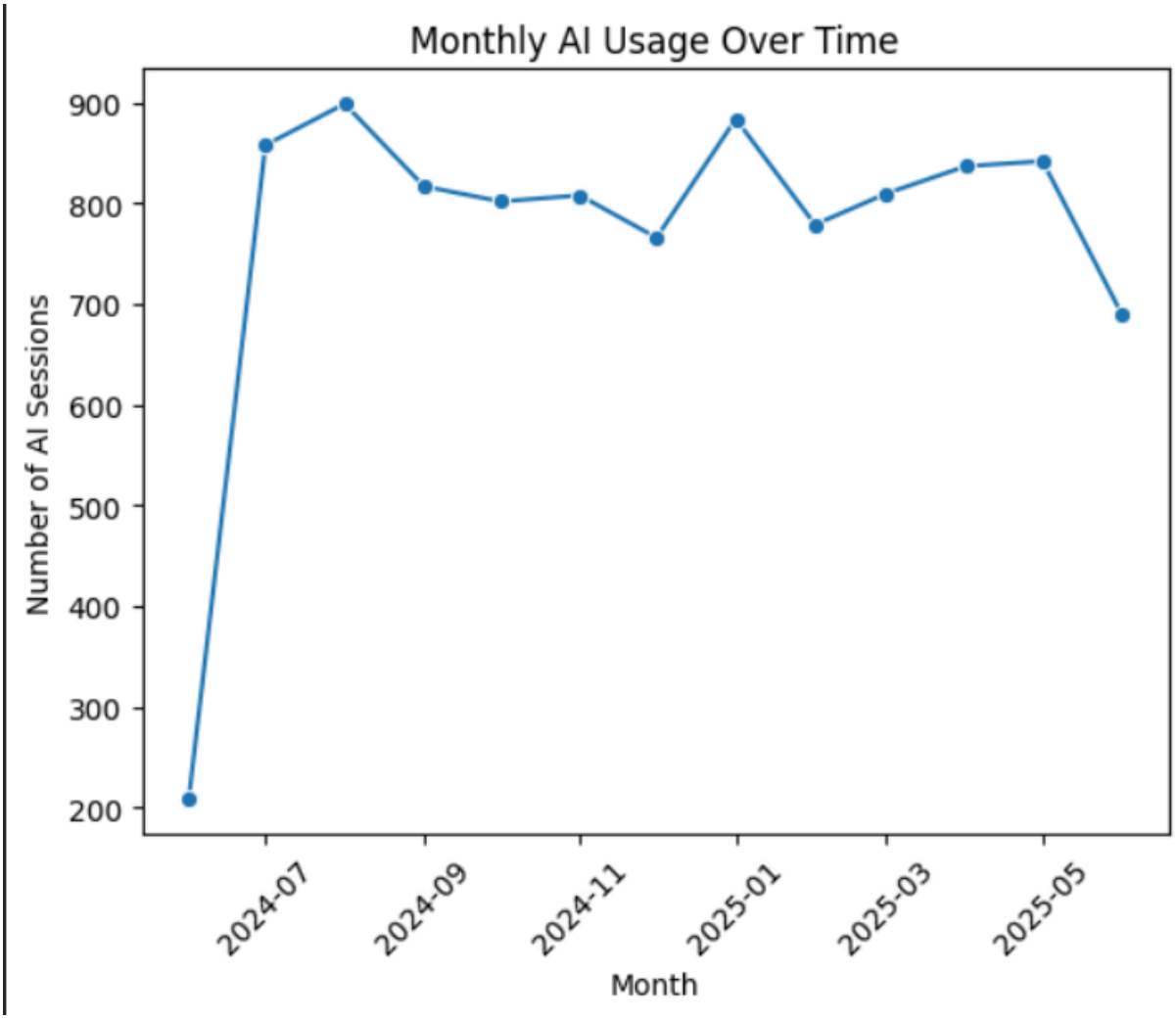
For research tasks, higher AI levels reduce the number of prompts, which means students get useful answers faster.

Coding and studying tasks show almost the same number of prompts at all AI levels, so AI level does not make much difference there. Overall, the impact of AI assistance level depends on the type of task.

8 Time Series Analysis

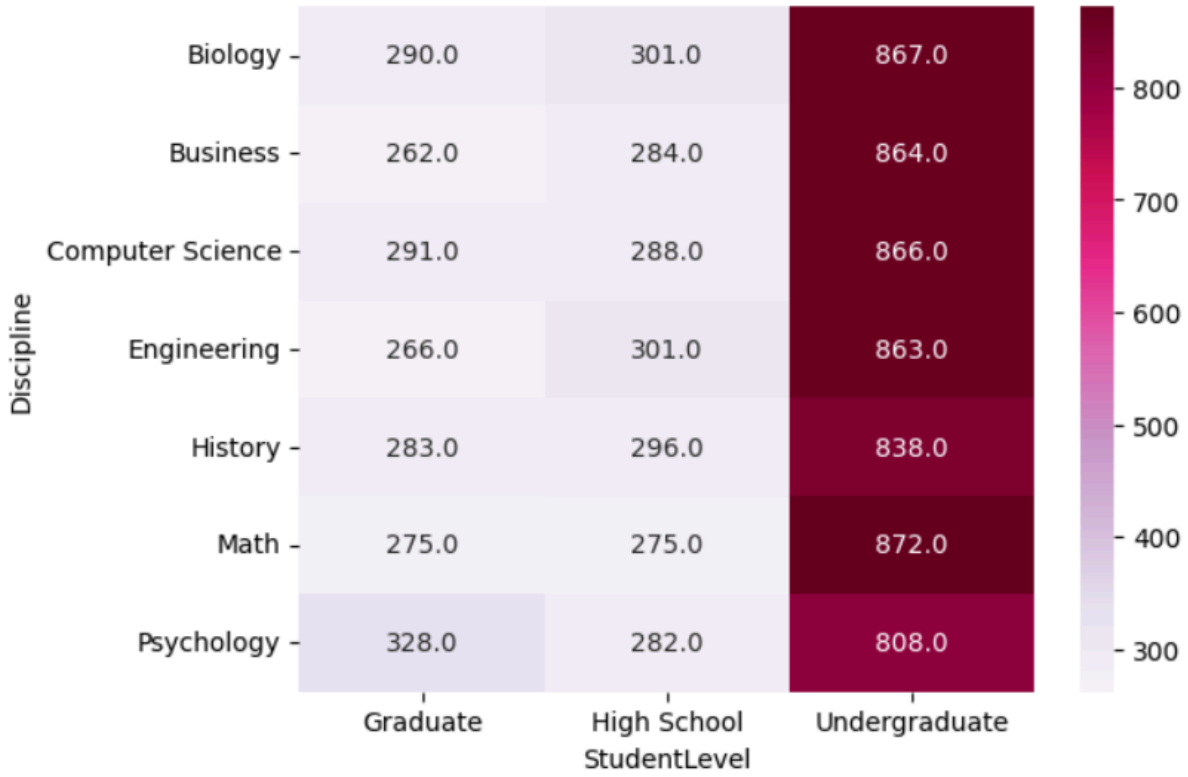
Here, in this dataset, for time series analysis, I extracted month, year, and year-month from the sessiondate column.

AI assistant usage by Year-Month



In this line plot, we can see that AI usage was highest in mid-2024 and early-2025. After mid-2025, the usage gradually decreased, maybe because exams were over.

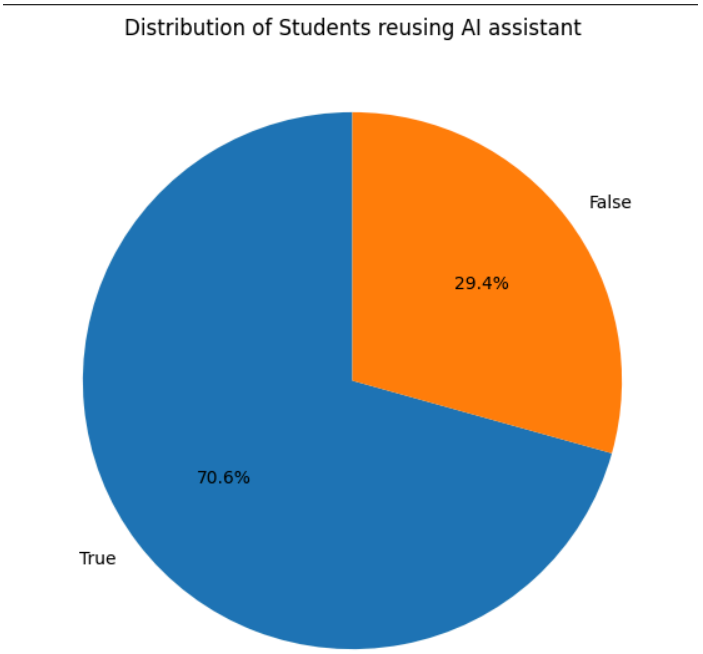
9 Relationship between Student Education and Course of students.



Mostly, undergraduate students from Maths use AI assistants the most. High school–level students from Engineering and Biology use AI tools at an average level.Business graduates use AI assistants very less.

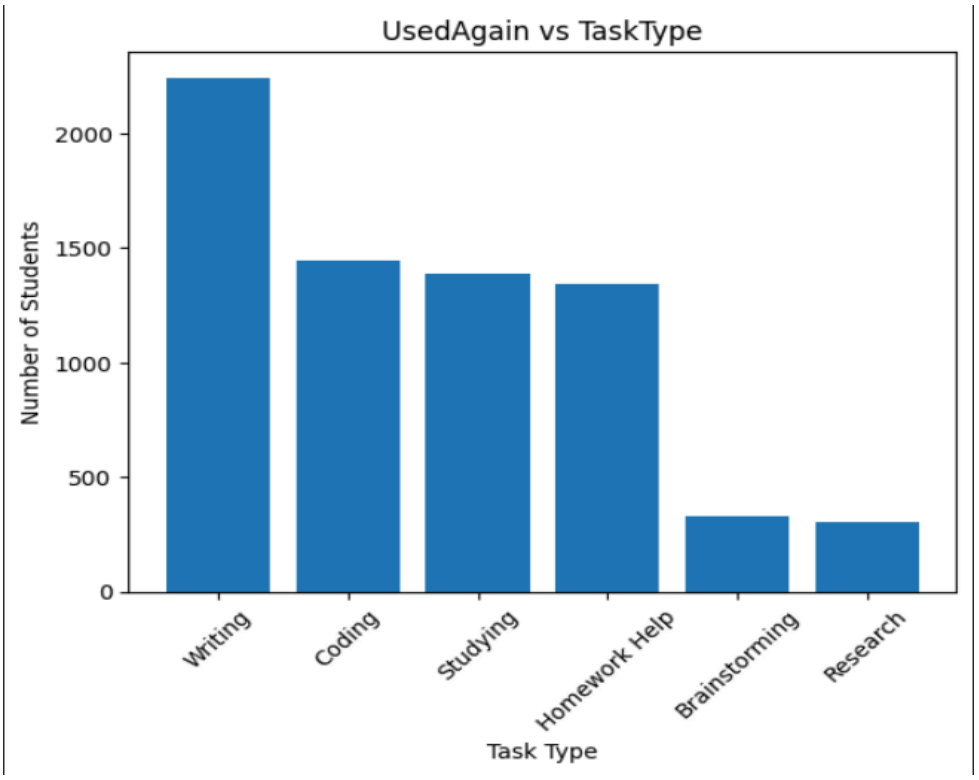
10 Distribution of Students reusing AI assistant

A positive response means that students are likely to use the AI assistant again.



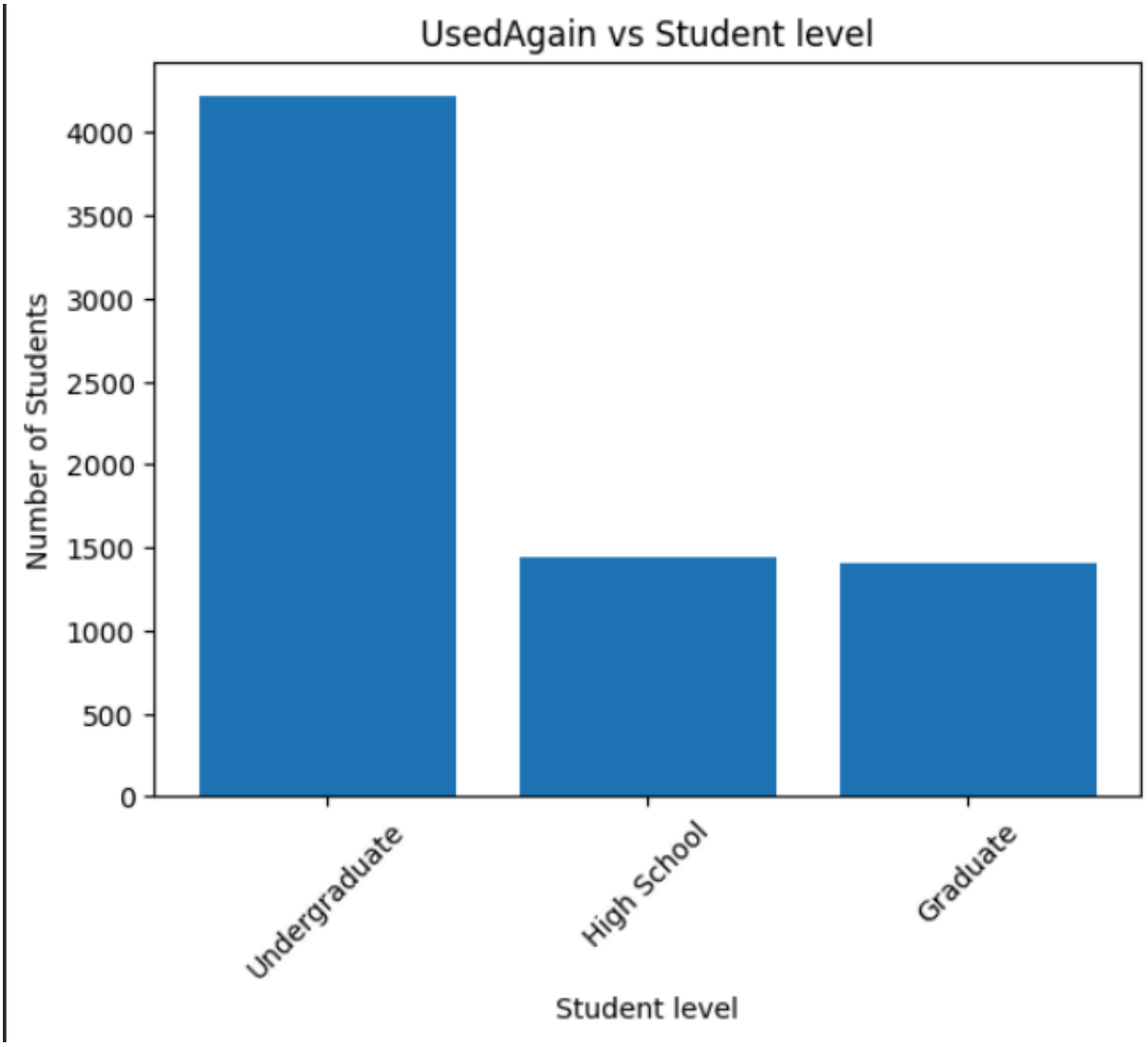
There is approximately 70% chances that students will use that AI tool again. It shows positive acceptance of AI tools by students.

11 UsedAgain vs Task Type.

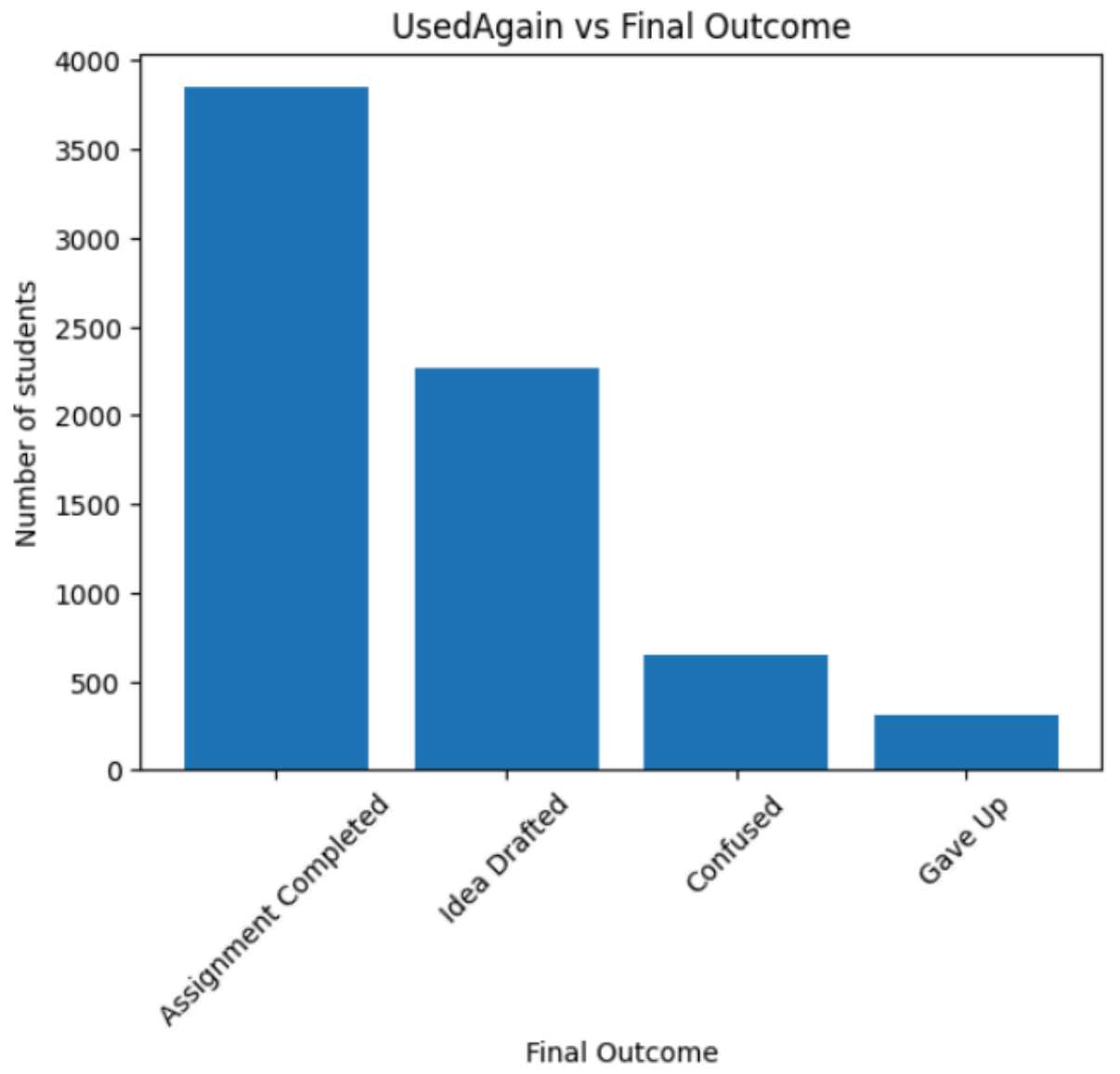


From this plot, we can see that students are more likely to use the AI assistant again for writing and assignment-related tasks.

12 UsedAgain vs student level

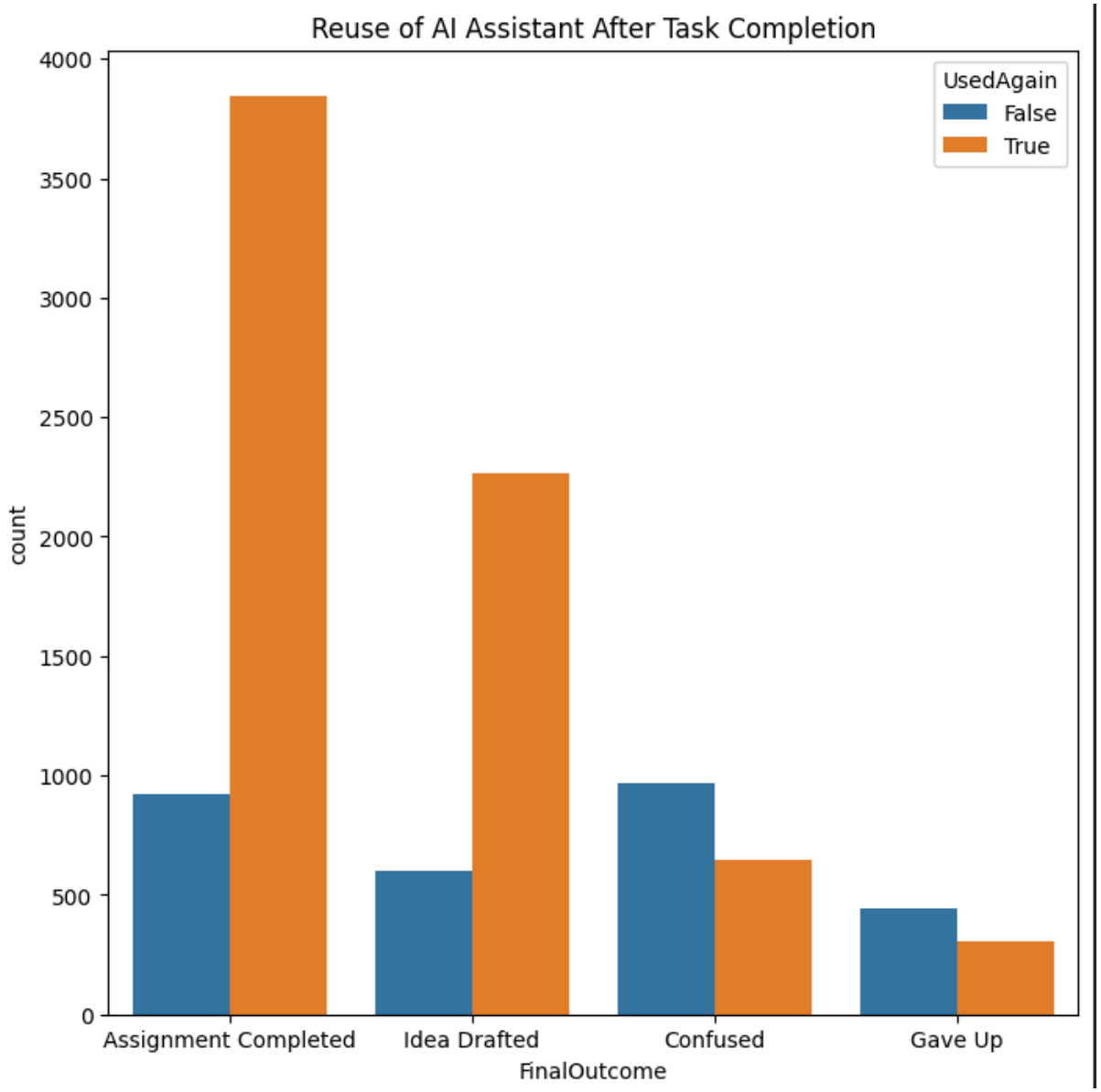


13. Used Again vs Final Outcome



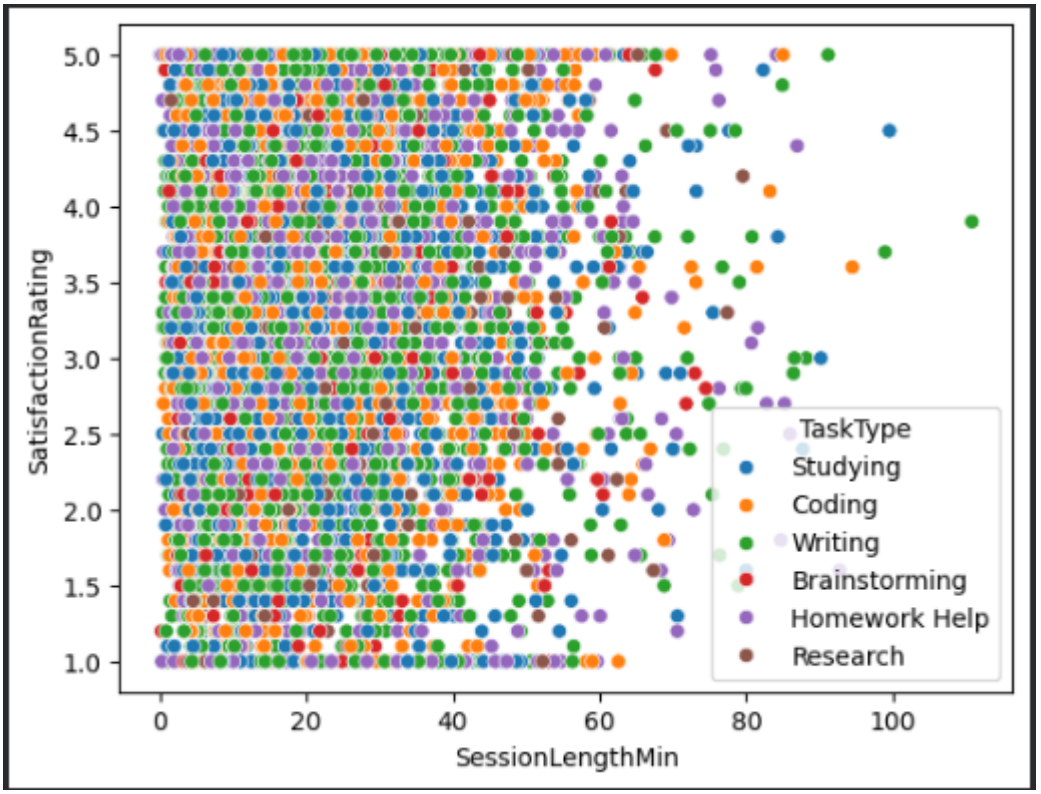
From these comparisons I observed that Undergraduate students Use Assistant for writing related assignment completion Tasks.

14. Students after finishing their task does they use same AI assistant again or not?



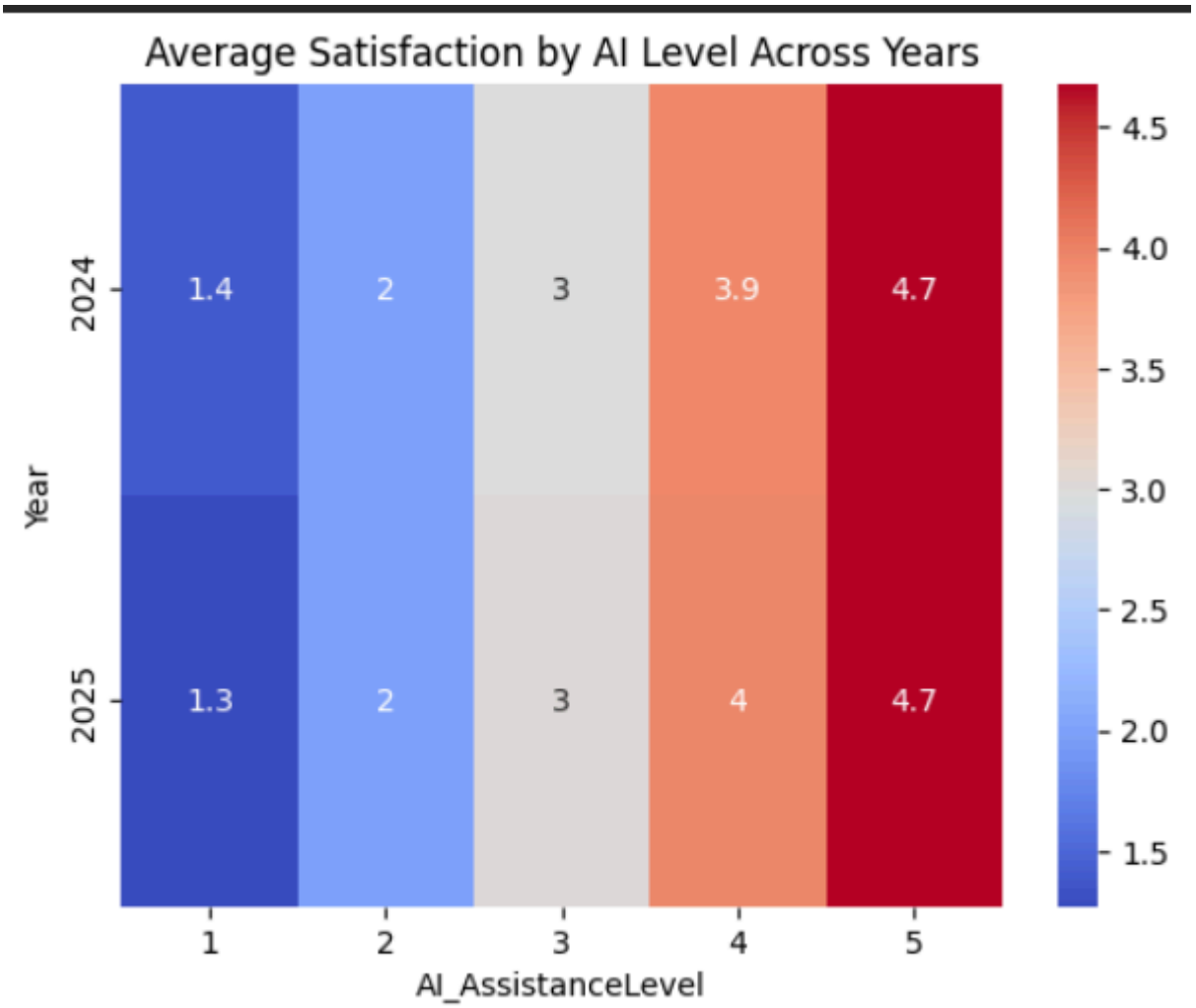
For Assignment Completed and Idea Drafted students mostly reuse that same AI assistant level.

15. How Session length is related to Satisfaction rating ?



As Session length in minutes increasing Satisfaction rating is also increasing. It indicates positive acceptance from student side of ai assistant tool.

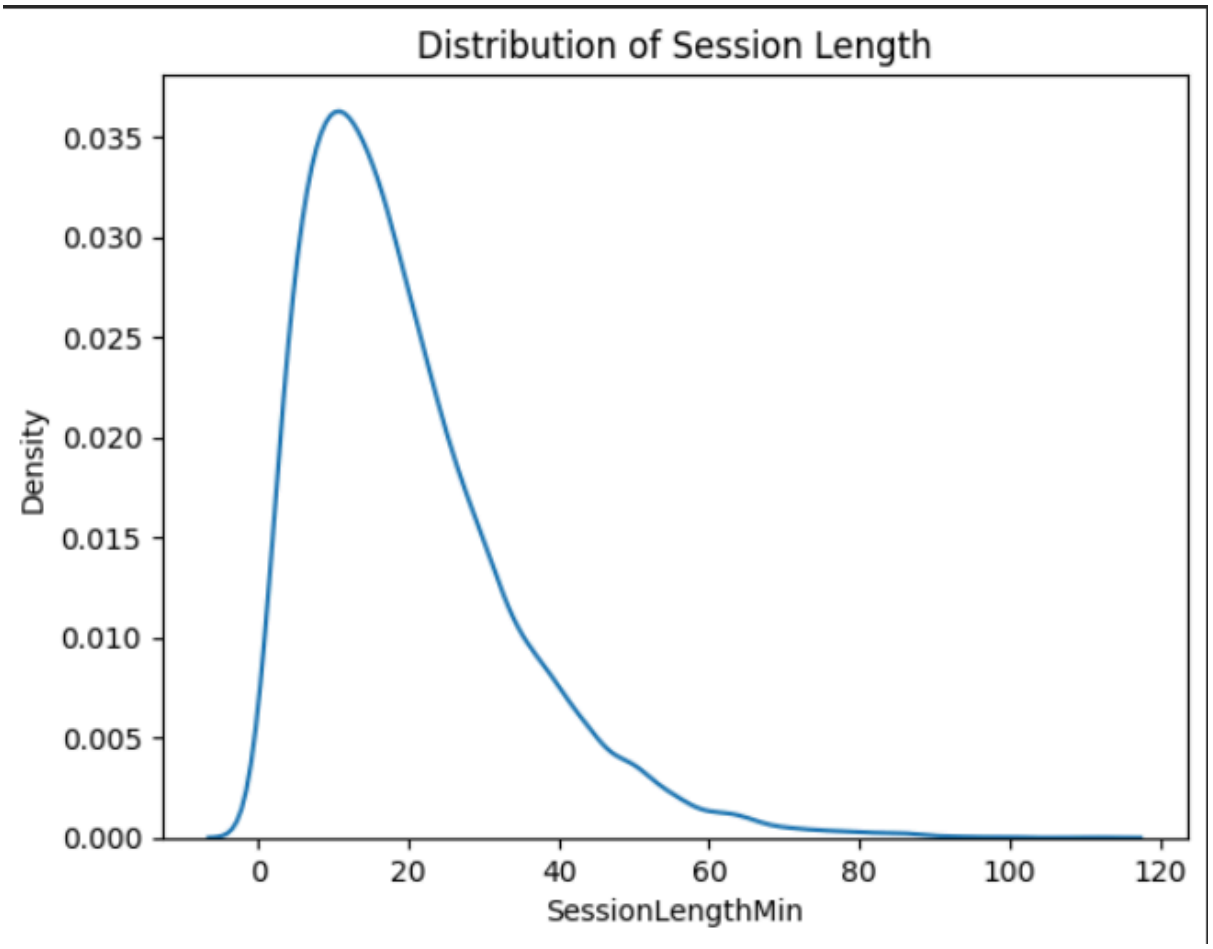
16 How Year by year ai assistants improved by satisfaction ratings of students.



This heatmap indicates performance of AI Assistance level is decrease by 0.1 .

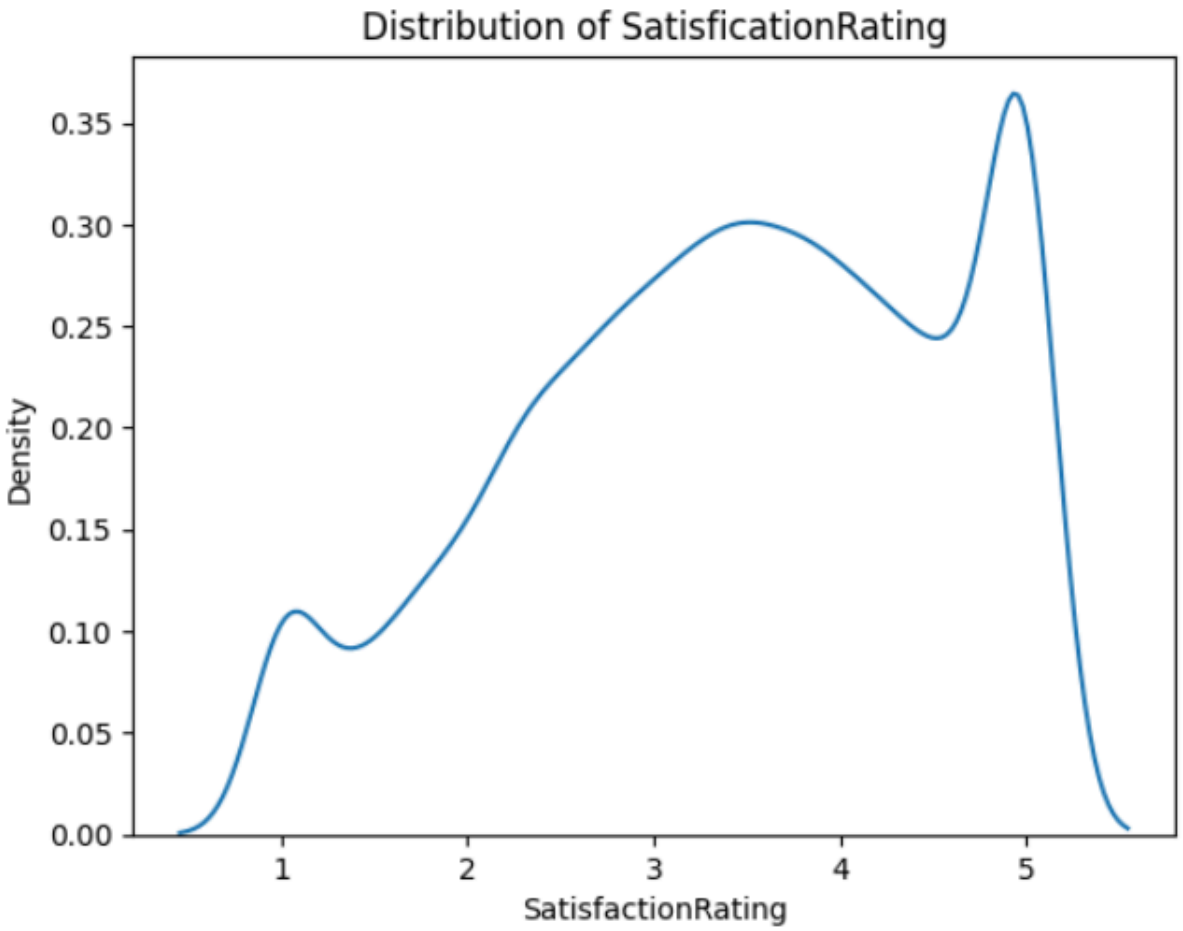
Performance of AI assistance level 2,3 and 5 remains constant in 2024 and 2025.

17 Distribution of session length



Mostly Interactions between AI assistant and Student are 1 to 55 minutes.

18. **Distribution of SatisfactionRating**



Most students rate high satisfaction with AI assistants, while a few students rate moderate satisfaction. Low satisfaction ratings are less, overall positive user experience.